Section A

For each question there are four possible answers, A, B, C and D. Choose the one you consider to be correct.

1 The relative atomic mass of boron, which consists of the isotopes $^{10}_{5}B$ and $^{11}_{5}B$ is 10.8.

What is the percentage of $^{11}_{5}B$ atoms in the isotopic mixture?

A 80 %
B 8.0 %
C 20 %
D 0.8 %

2 The use of the Data booklet is relevant to this question.

Many audio tapes are coated with iron (III) oxide. To find the mass of iron (III) oxide on a tape, the iron (III) oxide is quantitatively converted into $Fe^{2+}_{(aq)}$ and the solution is titrated against an appropriate reagent.

Which reagent could be used for this titration?

A $Sn^{4+}_{(aq)}$
B $S_4O_6^{2-}{}_{(aq)}$
C acidified $Cr_2O_7^{2-}$
D aqueous I_2

3 Which equation is used to define the first ionisation of bromine?

A $Br_{(g)} \rightarrow Br_{(g)}^{+} + e^{-}$

B $\frac{1}{2}Br_{2(g)} \rightarrow Br^{-}{}_{(g)} - e^{-}$

C $Br_{(g)} \rightarrow Br_{(g)}^{+} - e^{-}$

D $\frac{1}{2}Br_{2(g)} \rightarrow Br^{+}{}_{(g)} + e^{-}$

4 Which electronic configuration represents an element that forms an ion with a charge of –3 ?

A $1s^2 2s^2 2p^6 3s^2 3p^1$

B $1s^2 2s^2 2p^6 3s^2 3p^3$

C $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$

D $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$

5 What is observed when aqueous bromine is added to a solution of Cl^-, Br^- and I^- ions?

A a greenish vapour
B a red-brown vapour
C a pale yellow vapour
D a purple vapour

6 Which chloride has a giant ionic structure?

A Al_2Cl_6

B $SiCl_4$

C $MgCl_2$

D PCl_3

7 The first 8 successive ionisation energies of element Z in KJ mol^{-1} are 790, 1 600, 3 200, 4 400, 16 100, 19 800, 23 800 and 29 200.

Which group is element Z in?

A I

B II

C III

D IV

[Turn over

8 Which compound is a linear molecule?

A H_2O

B HCN

C SO_2

D Cl_2O

9 Which molecule contains six bonding electrons?

A NCl_3

B CO_2

C H_2S

D C_2H_4

10 Ethanol is much more soluble in water than ethylethanoate.

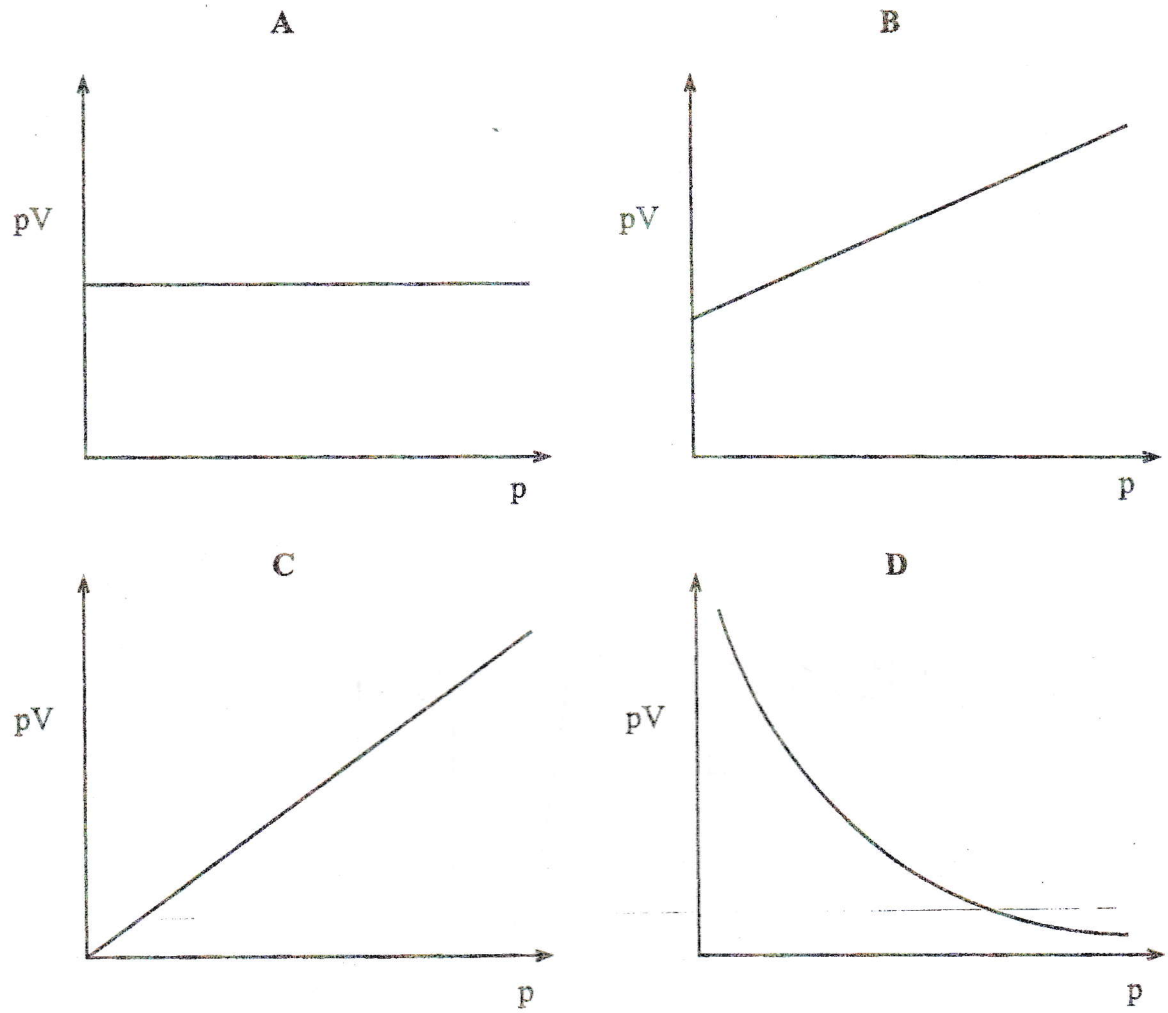
Which statement correctly accounts for this observation?

A A hydrogen bond forms between the hydrogen of the – OH group in ethanol and the oxygen of a water molecule.
B Ethanol is a non-polar molecule but ethylethanoate is polar.
C A hydrogen bond forms between the hydrogen of the – OH group in ethanol and the hydrogen of a water molecule.
D Ethanol is a polar molecule but ethylethanoate is non-polar.

11 Nitrates of Group (III) elements decompose when heated to produce

A a mixture of nitrogen dioxide and oxygen.
B the oxide of the metal and oxygen.
C the nitrite of the metal and nitrogen dioxide.
D the oxide and the nitrite of the metal.

12 Which curve shows the correct graph of pV against p for a fixed mass of an ideal gas at constant temperature?



[Turn over

13 The enthalpy changes of formation of ethene and ethane are +52 KJ mol $^{-1}$ and – 85 KJmol $^{-1}$ respectively at 298 K.

What is the enthalpy change of reaction at 298 K for the following reaction?

$C_2H_{4(g)} + H_{2(g)} \rightarrow C_2H_{6(g)}$

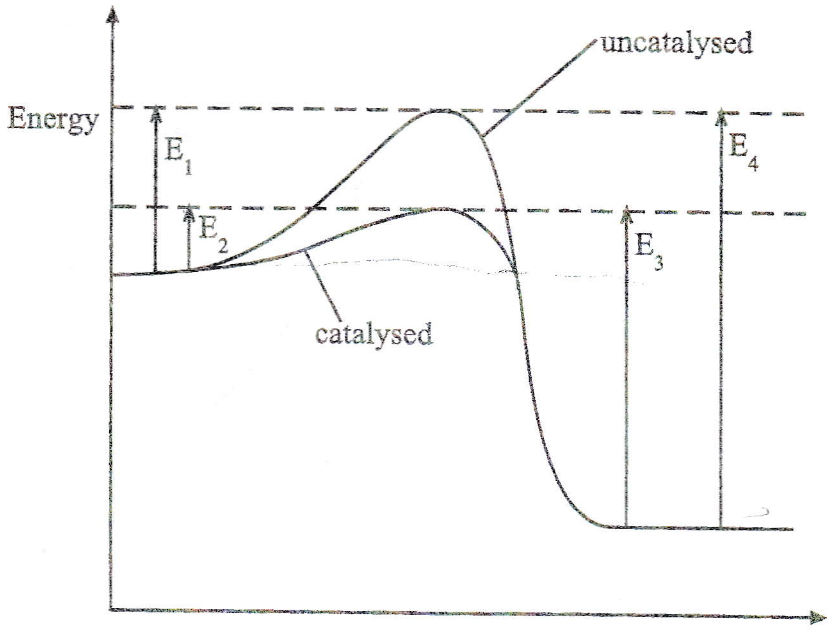
A – 137 KJmol $^{-1}$

B –33 KJmol $^{-1}$

C +33 KJmol $^{-1}$

D +137 KJmol $^{-1}$

14 The energy diagram represents the reaction occurring with and without a catalyst.



Which statement is correct?

A E_4 is the activation energy for the reverse catalysed reaction.
B The forward reaction, with a catalyst, is exothermic.
C The enthalpy change of reaction is reduced by using a catalyst.
D The enthalpy change of reaction is $E_2 - E_3$.

15 The use of the Data booklet is relevant to this question.

In acidic solution, MnO_4^- ions oxidise Cl^- ions to Cl_2. The value of E^θ for the reaction is +0.16V.

Which statement is correct?

A The oxidation number of chlorine changes from – 1 to + 2 .
B The oxidation number of manganese changes from +7 to + 4.
C The oxidation number of manganese changes form +7 to + 6.
D The oxidation number of chlorine changes from – 1 to 0.

16 Which property of Group (IV) oxides is true?

A all the oxides are acidic
B acidity increases down the group
C two of the oxides are acidic
D only one of the oxides is amphoteric

17 The pK_b value for aqueous ammonia at 25 °C is 4. 8.

What is the correct pK_a value for the ammonium ion at this temperature?

A 2.4

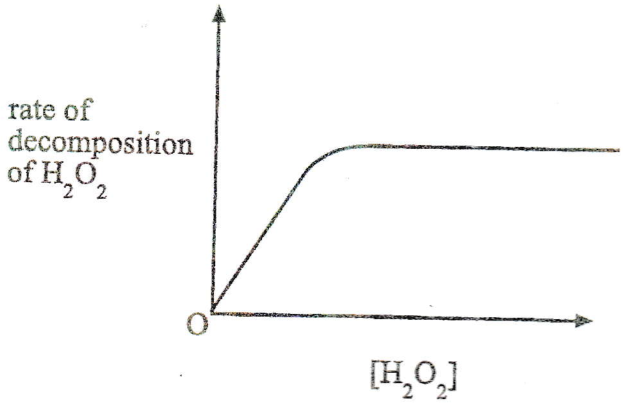
B 3.4

C 4.8

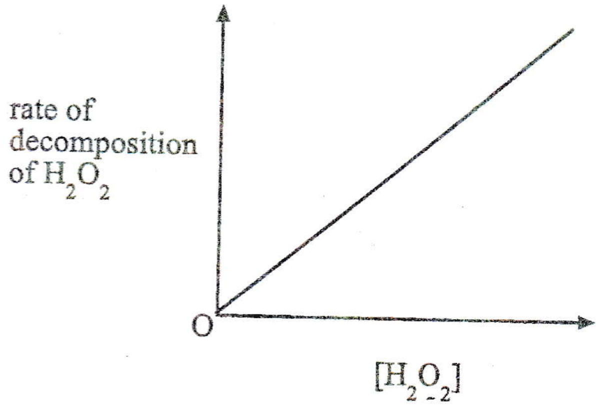
D 9.2

18 Which graph would confirm that the decomposition of hydrogen peroxide is first order with respect to the concentration of hydrogen peroxide?

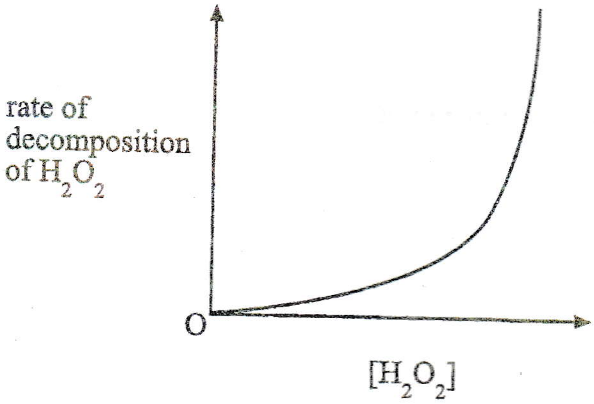
A



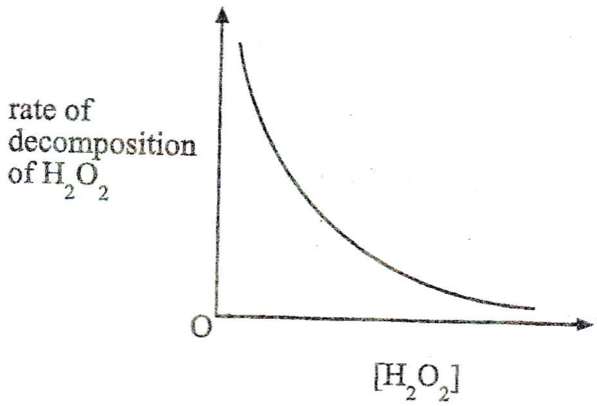
B



C



D



19 The table gives data for the reaction between **X** and **Y** at constant temperature.

Experiment	[X] /moldm $^{-3}$	[Y] /moldm $^{-3}$	Initial rate /moldm $^{-3}$s $^{-1}$
1	0.3	0.2	4.0×10^{-4}
2	0.6	0.4	1.6×10^{-3}
3	0.6	0.8	6.4×10^{-3}

The rate equation for the reaction is

A $k[X][Y]^2$.

B $k[X]^2[Y]$.

C $k[X]^2$.

D $k[Y]^2$.

20 Which statement is most likely to be true for astatine, an element below iodine in Group (VII) of the Periodic Table?

A Sodium astatide and hot concentrated sulphuric acid react to form astatine.
B Potassium astatide and hot dilute sulphuric acid react to form white fumes of hydrogen astatide only.
C Silver astatide reacts with dilute aqueous ammonia in excess to form a solution of a soluble complex.
D Astatine and aqueous potassium chloride react to form aqueous potassium astatide and chlorine.

21 Which atom has three unpaired electrons?

A Al

B Sc

C Cr

D Co

22 The use of Data booklet is relevant to this question.

Which aqueous solution should **not** be stirred with a nickel spatula?

A $Fe^{3+}(aq)$

B $Co^{2+}(aq)$

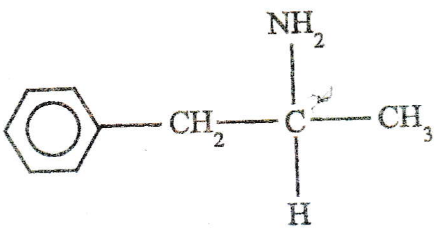
C $Cr^{2+}(aq)$

D $Mn^{2+}(aq)$

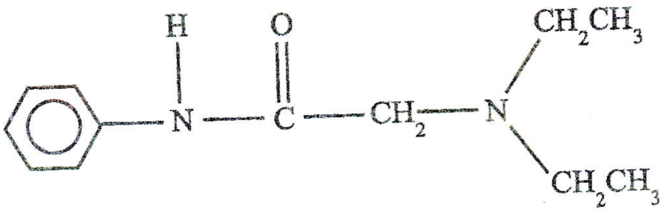
23 Which statement always applies to a nucleophile?

A it is negatively charged
B it is a single atom
C it has a lone pair of electrons
D it attacks a double bond

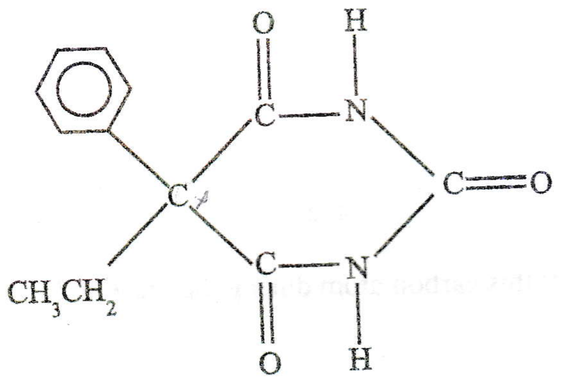
24 The diagrams show the structure of three drugs.



amphetamine



lidocaine



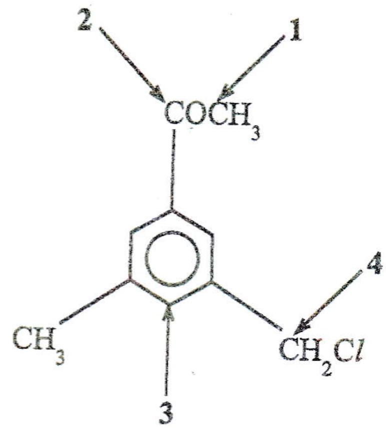
phenobarbital

What is the total number of chiral centres in these three structures?

A 0
B 1
C 2
D 3

[Turn over

25 At which carbon atom in the molecule shown is electrophilic attack most likely?



A 1
B 2
C 3
D 4

26 During the nitration of benzene, a nitro group substitutes at a carbon atom.

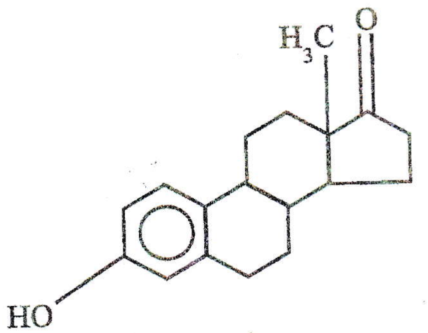
Which option gives the arrangement of the bonds at this carbon atom during the reaction?

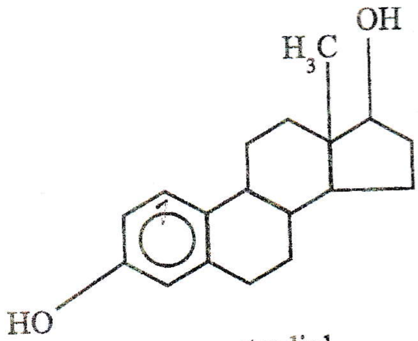
	At the start of the reaction	**In the intermediate complex**	**At the end of the reaction**
A	planar	planar	planar
B	planar	tetrahedral	planar
C	planar	tetrahedral	tetrahedral
D	tetrahedral	planar	tetrahedral

27 Which reagent reacts with trichloroethene to give a chiral product?

A Br_2
B H_2
C HCl
D $NaOH_{(aq)}$

28 Two female sex hormones are oestrone and oestradiol


oestrone


oestradiol

Which reagent could be used to distinguish between the hormones?

A $Br_{2(aq)}$ at room temperature
B 2, 4 – dinitrophenylhydrazine
C Fehling's reagent
D I_2 in $NaOH_{(aq)}$

29 With which reagent do phenylamine and phenol have similar reactions?

A $Br_{2(aq)}$
B HCl
C $HNO_{2\,(aq)}$
D $NaOH_{(aq)}$

[Turn over

30 Which compound gives

(i) fumes of HCl with PCl_5 and

(ii) NH_3 when heated with $NaOH_{(aq)}$

A $HOCH_2CH_2NH_2$

B $HOCH_2CONH_2$

C $HOCH_2C(NH_2)HCO_2H$

D $NH_2CH_2CO_2H$

SECTION B

For each of the questions in this section. one or more of the three numbered statements **1** *to* **3** *may be correct.*

Decide whether each of the statements is or is not correct. (You may find it helpful to put a tick against the statement(s) which you consider to be correct).

The responses **A** *to* **D** *should be selected on the basis of*

A	B	C	D
1,2 and 3 are correct	*1 and 2 only are correct*	*2 and 3 only are correct*	*1 only is correct*

No other combination of statements is used as a correct response.

31 For which of the following reactions does the value of ΔH^θ represent both a standard enthalpy change of combustion and a standard enthalpy change of formation?

1 $C_{(S)} + O_{2(g)} \rightarrow CO_{2(g)}$

2 $2C_{(s)} + O_{2(g)} \rightarrow 2CO_{(g)}$

3 $CO_{(g)} + \frac{1}{2}O_{(g)} \rightarrow CO_{2(g)}$

32 An account in a student's notebook read:

"...an excess of aqueous bromine was added to aqueous phenol in a test tube and 2,4,6 – tribromophenol was produced as a white precipitate suspended in a yellow alkaline solution..."

Which statement(s) about this account is/are **not** correct?

1 The resultant solution is alkaline.

2 The precipitate is 2, 4, 6 – tribromophenol.

3 The precipitate formed was white.

[Turn over

33 If the reaction:

$$X + Y \rightarrow W + Z,$$

is described as being zero order with respect to X, it means that

1 the rate of reaction is independent of the concentration of X.

2 the concentration of X does not change during the reaction.

3 X molecules do not possess sufficient energy to react.

34 A current is produced in the following cell.

$$Pt|Y^{2+}_{(aq)}, Y^{3+}_{(aq)}||XO^-_{4(aq)}, X^{2+}_{(aq)}, |Pt$$

Electrode	$E^\theta|298\,K$
$Y^{3+}_{(aq)}|Y^{2+}_{(aq)}$	+0.77V
$XO^-_{4(aq)}|X^{2+}_{(aq)}$	+1.52V

Which is correct about the cell that is formed?

1 E^θ cell is + 0.75 V.

2 Y^{3+} is being reduced to Y^{2+}.

3 X^{2+} is being oxidised to XO_4^-.

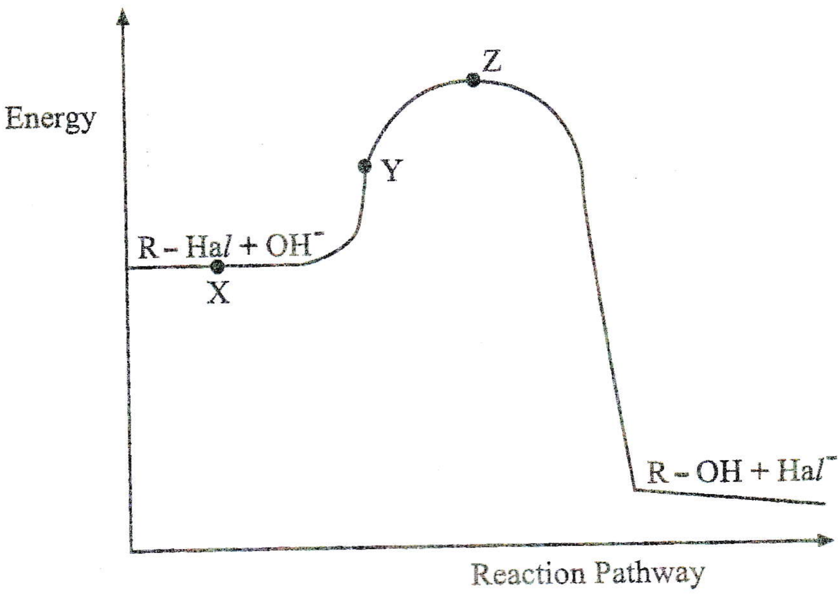
35 Which substance(s) is/are **not** oxidised by hot acidified aqueous potassium manganate (VII)?

1 ethanoic acid

2 ethanol

3 ethane – 1, 2 - diol

36 The graph shows an energy profile for a reaction between a halogenoalkane and an aqueous alkali.



Which statement is /are **not** correct?

1 Energy difference between X and Y represents the activation energy.

2 The reaction is an example of nucleophilic substitution.

3 The C – Hal bond weakens between X and Z.

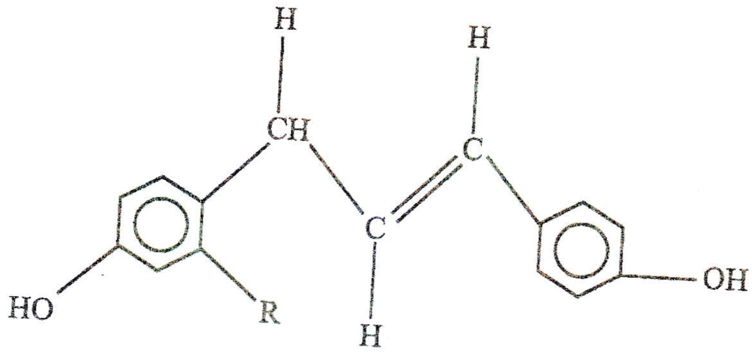
37 Which type of reaction(s) does the organic compound shown undergo?

1 free-radical substitution

2 nucleophilic substitution

3 reduction

38 The compound shown is secreted by a parasitic plant to enable it to recognise its host before settling and growing on it.



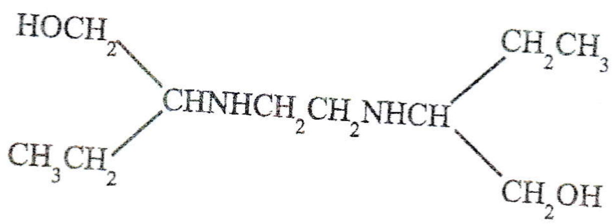
Which reagent(s) would react with this compound?

1 aqueous sodium carbonate

2 aqueous bromine

3 aqueous sodium hydroxide

39 Which reagent(s), under suitable conditions, could convert ethambutol into an organic compound containing a halogen?



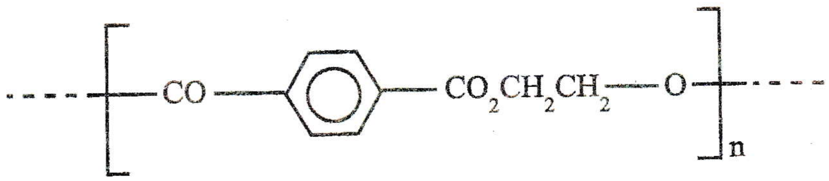
ethambutol

1 hydrogen bromide

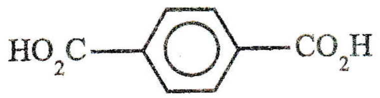
2 alkaline aqueous iodine

3 ethanoyl chloride.

40 Terylene is formed by condensation polymerisation and contains the repeat unit shown:



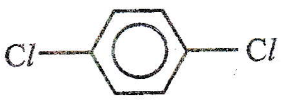
From which pair(s) of starting materials could it be formed?

1. 

and $HOCH_2CH_2OH$

2. 

and $HOCH_2CH_2OH$

3. 

and $HO_2CCH_2CH_2CO_2H$